

Phishing Detection & Awareness Report

1. Introduction

What is Phishing?

Phishing is a type of cyberattack in which attackers impersonate trusted individuals, organizations, or services to deceive users into revealing sensitive information or performing actions that compromise security. These attacks are commonly delivered through emails but can also occur via text messages (SMS phishing or smishing), phone calls (voice phishing or vishing), social media platforms, and fake websites.

The primary objective of phishing is to manipulate users into:

- Clicking malicious links
- Downloading malware-infected attachments
- Entering usernames and passwords on fake login pages
- Sharing confidential information such as banking details, credit card numbers, or One-Time Passwords (OTPs)

Phishing attacks rely on social engineering techniques that exploit human trust rather than technical vulnerabilities, making them one of the most effective methods used by cybercriminals.

Why is Phishing Dangerous?

Phishing is considered one of the most serious cybersecurity threats because it can lead to significant financial, operational, and reputational damage. Even a single successful phishing attack can compromise an entire organization's network.

Some major risks associated with phishing include:

- **Data Breaches:** Attackers can steal confidential business information, customer records, and intellectual property.
- **Credential Theft:** Stolen usernames and passwords may be used to gain unauthorized access to corporate systems and cloud services.
- **Financial Loss:** Employees may unknowingly transfer money or disclose banking information to cybercriminals.
- **Malware and Ransomware Infections:** Phishing emails often contain malicious attachments or links that install malware or ransomware on the victim's device.
- **Identity Theft:** Personal information obtained through phishing can be used to impersonate individuals or commit fraud.
- **Business Disruption:** Successful phishing attacks can interrupt business operations, reduce productivity, and require costly incident response efforts.
- **Reputational Damage:** Organizations that suffer phishing-related breaches may lose customer trust and face legal or regulatory consequences.

According to numerous cybersecurity reports, phishing remains one of the leading causes of security incidents worldwide because it targets human behavior rather than software vulnerabilities.

Why Organizations Need Phishing Awareness

Technology alone cannot prevent every phishing attack. Firewalls, antivirus software, email security gateways, and intrusion detection systems provide important layers of protection, but attackers continually develop new techniques that bypass technical controls.

Employees are often the first line of defense against phishing attacks. Therefore, organizations must invest in phishing awareness and cybersecurity training to help users recognize suspicious emails and respond appropriately.

Phishing awareness programs enable employees to:

- Identify suspicious emails, links, and attachments.
- Verify the authenticity of unexpected requests.
- Recognize common phishing indicators such as urgent language, spelling errors, spoofed email addresses, and fake login pages.
- Report suspected phishing attempts to the organization's security team.
- Avoid actions that could compromise sensitive information or organizational systems.

Regular phishing awareness training significantly reduces the likelihood of successful attacks by improving employees' ability to detect and respond to phishing attempts. Creating a strong security-aware culture helps organizations protect sensitive data, maintain business continuity, and reduce the financial and operational impact of cyber threats.

2. Objectives

Example:

- Analyze phishing email samples.
- Identify phishing indicators.
- Classify email risks.
- Explain phishing attacks in simple language.
- Provide awareness and prevention guidelines.

3. Tools Used

Tool	Purpose
Google Message Header Analyzer	Analyze email headers
MXToolbox Email Header Analyzer	Verify sender information
Web Browser	Inspect URLs and domains safely
Microsoft Word / Google Docs / Canva	Report writing
GitHub	Documentation and version control

4. Phishing Email Samples

Note : All email samples used in this report were created and sent by me between my own email accounts in a controlled environment for educational and cybersecurity awareness purposes. Three emails were intentionally designed to simulate phishing attacks, while one email represents a legitimate message for comparison.

This section presents three phishing email examples and one legitimate email example. Each sample is analyzed by reviewing the sender information, subject line, email content, and initial observations that help identify whether the email is malicious or genuine.

Sample 1 – Credential Theft Phishing Email

Subject

Urgent: Your Microsoft 365 Account Will Be Suspended Today

Email Body

Dear User,

We detected unusual activity on your Microsoft 365 account.

Your account will be permanently suspended within 24 hours unless you verify your identity immediately.

Click the link below to verify your account:

<https://microsoft365-secure-login.com>

Failure to verify your account may result in permanent data loss.

Thank you,

Microsoft Security Team

Urgent: Your Microsoft 365 Account Will Be Suspended Today Inbox x



Trial <trialcloud11@gmail.com>
to me ▾

15:36 (23 minutes ago) ☆ ☺ ↶ ⋮

Dear User,

We detected unusual activity on your Microsoft 365 account.

Your account will be permanently suspended within 24 hours unless you verify your identity immediately.

Click the link below to verify your account:

<https://microsoft365-secure-login.com>

Failure to verify your account may result in permanent data loss.

Thank you,

Microsoft Security Team

↶ Reply ↷ Forward ☺

MessageId	CA0J4DvonS_W-aM1GmKnVqQ5pr-XQMylfkuo+xepAqML4=ME1Bw@mail.gmail.com
Created at:	7/7/2026, 3:36:28 PM GMT+5:30 (Delivered after 18 sec)
From:	Trial <trialcloud11@gmail.com>
To:	"shlokcloud1130@gmail.com" <shlokcloud1130@gmail.com>
Subject:	Urgent: Your Microsoft 365 Account Will Be Suspended Today
DKIM:	pass with domain gmail.com Learn more

#	Delay	From *	To *	Protocol	Time received
0	17 sec		→ 2002:a05:6402:5296:b0:698:3f5c:1b84	SMTP	7/7/2026, 3:36:45 PM GMT+5:30
1		mail-sor-f41.google.com.	→ [Google] mx.google.com		7/7/2026, 3:36:45 PM GMT+5:30 <i>Originated at Gmail</i>
2			→ [Google] 2002:a05:6402:3225:b0:698:4405:4feb	SMTP	7/7/2026, 3:36:45 PM GMT+5:30
3	1 sec		→ [Google] 2002:ac0:ee1c:0:b0:3f4:5bce:1c6c	SMTP	7/7/2026, 3:36:46 PM GMT+5:30

Initial Observations

- The sender's email address is not an official Microsoft domain.
- Creates a false sense of urgency.
- Threatens account suspension.
- Contains a suspicious login link.
- Uses fear to encourage immediate action.
- Attempts to steal Microsoft login credentials.

Sample 2 – Fake Bank Phishing Email

Subject

Important Security Alert – Verify Your Bank Account

Email Body

Dear Customer,

Your online banking account has been temporarily restricted due to suspicious transactions.

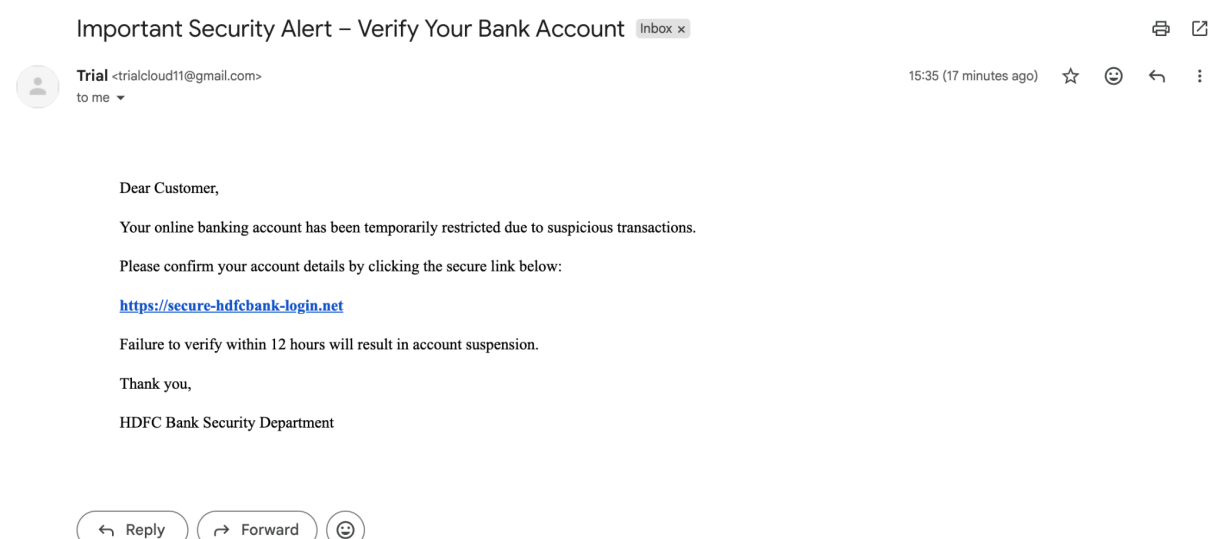
Please confirm your account details by clicking the secure link below:

<https://secure-hdfcbank-login.net>

Failure to verify within 12 hours will result in account suspension.

Thank you,

HDFC Bank Security Department



MessageId	CAOJ4DvqmV1wT7WYp6GjDFi86q_faweFgp9-6-ceegzmPB2gU8g@mail.gmail.com
Created at:	7/7/2026, 3:35:40 PM GMT+5:30 (Delivered after 13 sec)
From:	Trial <trialcloud11@gmail.com>
To:	'shlokcloud1130@gmail.com' <shlokcloud1130@gmail.com>
Subject:	Important Security Alert – Verify Your Bank Account
DKIM:	pass with domain gmail.com Learn more

#	Delay	From *	To *	Protocol	Time received
0	11 sec		→ 2002:a17:906:1809:b0:c12:33a2:a634	SMTP	7/7/2026, 3:35:51 PM GMT+5:30
1	2 sec	mail-sor-f41.google.com.	→ [Google] mx.google.com		7/7/2026, 3:35:53 PM GMT+5:30 <i>Originated at Gmail</i>
2			→ [Google] 2002:a17:907:da10:b0:c12:701b:e531	SMTP	7/7/2026, 3:35:53 PM GMT+5:30
3			→ [Google] 2002:ac0:ee1c:0:b0:3f4:5bce:1c6c	SMTP	7/7/2026, 3:35:53 PM GMT+5:30

Initial Observations

- The sender is not using the official bank domain.
- Fake website URL attempts to imitate the bank.
- Uses urgency and fear.
- Requests sensitive banking information.
- Generic greeting instead of the customer's name.
- Strong indication of a phishing attempt.

Sample 3 – Fake Invoice Email (Malware Delivery)

Subject

Invoice INV-45892 – Payment Due

Email Body

Hello,

Please find attached your pending invoice INV-45892.

Kindly review the attached document and complete the payment today.

Attachment:

Invoice_INV45892.zip

Regards,

Accounts Department

Invoice INV-45892 – Payment Due



Trial <trialcloud11@gmail.com>
to shlokcloud1130@gmail.com ▾

Hello,

Please find attached your pending invoice INV-45892.

Kindly review the attached document and complete the payment today.

Attachment:

Invoice_INV45892.zip

Regards,

Accounts Department

↩ Reply

➦ Forward



Initial Observations

- Unexpected invoice from an unknown sender.
- ZIP attachment may contain malware.
- No company name is mentioned.
- Pressures the recipient to act quickly.
- Generic email without transaction details.
- High risk of malware infection if the attachment is opened.

Sample 4 – Legitimate Email

Subject

Monthly Team Meeting – July Schedule

Email Body

Dear Team,

This is a reminder that our monthly team meeting is scheduled for Friday at 10:00 AM in Conference Room A.

The meeting agenda has been attached for your reference.

If you have any questions, please contact the HR department.

Thank you.

Human Resources

Monthly Team Meeting – July Schedule



Trial <trialcloud11@gmail.com>
to shlokcloud1130@gmail.com ▾

3:40 PM (3 hours ago)

Dear Team,

This is a reminder that our monthly team meeting is scheduled for Friday at 10:00 AM in Conference Room A.

The meeting agenda has been attached for your reference.

If you have any questions, please contact the HR department.

Thank you.

Human Resources

↩ Reply

➡ Forward

😊

Original Message

Message ID	<CAOJ4DvoFCN06QWv7+ZAw2=9Td1ohQN6k7ToqHT-jZsdWkc_vPw@mail.gmail.com>
Created at:	Tue, Jul 7, 2026 at 3:40 PM (Delivered after 0 seconds)
From:	Trial <trialcloud11@gmail.com>
To:	"shlokcloud1130@gmail.com" <shlokcloud1130@gmail.com>
Subject:	Monthly Team Meeting – July Schedule

[Download Original](#)



Initial Observations

- Sender uses the organization's official email domain.
- Professional language with no urgency or threats.
- Does not request passwords or confidential information.
- Contains relevant business information.
- Expected communication from the HR department.
- Safe and legitimate email.

3. Inspect Sender Domain & Links

The sender's email address and hyperlinks were examined to determine whether they belonged to the organization the email claimed to represent. A mismatch between the claimed organization and the actual domain is a common phishing indicator.

Sample 1 – Fake Microsoft Account Verification

Claimed Organization: Microsoft

Expected Official Domain: microsoft.com

Suspicious Link: <https://microsoft365-secure-login.com>

Observation: The sender domain does not belong to Microsoft's official domain (**microsoft.com**). The embedded link also redirects to a non-Microsoft website that imitates the Microsoft login page to steal user credentials.

Result: High Risk – Phishing Email

Sample 2 – Fake HDFC Bank Security Alert

Claimed Organization: HDFC Bank

Expected Official Domain: hdfcbank.com

Suspicious Link: <https://secure-hdfcbank-login.net>

Observation: The sender domain is different from the official HDFC Bank domain (**hdfcbank.com**). The email attempts to trick users into entering banking credentials on a fake website.

Result: High Risk – Phishing Email

Sample 3 – Fake Invoice Email

Claimed Organization: Accounts Department

Expected Official Domain: The sender should belong to the actual supplier or organization's official domain.

Attachment: Invoice_INV45892.zip

Observation: The sender uses an unknown external domain and includes a ZIP attachment, which could contain malware. The email does not identify the company and pressures the recipient to make an immediate payment.

Result: Critical Risk – Malware Delivery Attempt

Sample 4 – Legitimate HR Meeting Reminder

Claimed Organization: Human Resources Department

Expected Official Domain: company.com

Observation: The sender's email address matches the organization's official domain. The email contains routine business information, uses professional language, and does not request passwords, financial information, or immediate action.

Result: Legitimate Email

4. Identify Phishing Indicators

The following phishing indicators were identified during the analysis:

Sample 1 – Fake Microsoft Email

- Unofficial Microsoft sender domain.
- Urgent message claiming the account will be suspended.
- Suspicious login URL.
- Requests immediate account verification.
- Attempts to steal Microsoft account credentials.

Sample 2 – Fake HDFC Bank Email

- Sender domain does not match the official HDFC Bank domain.
- Generic greeting ("Dear Customer").

- Creates urgency by threatening account suspension.
- Contains a fake banking login link.
- Attempts to collect banking credentials.

Sample 3 – Fake Invoice Email

- Unexpected invoice from an unknown sender.
- ZIP attachment that may contain malware.
- No company identification or purchase details.
- Urges immediate payment.
- High possibility of malware infection.

Sample 4 – Legitimate HR Email

- Official company email domain.
- Professional and expected business communication.
- No suspicious links or attachments.
- Does not request confidential information.
- Normal meeting reminder with a clear business purpose.

5. Classify Email Risk

Email Sample	Risk Level	Reason
Fake Microsoft Account Verification	High	Attempts to steal Microsoft login credentials through a fake website.
Fake HDFC Bank Security Alert	High	Designed to steal online banking credentials using a fake login page.
Fake Invoice Email	Critical	Contains a suspicious ZIP attachment that may deliver malware.
Legitimate HR Meeting Reminder	Legitimate (Low Risk)	Official sender domain, expected business communication, and no phishing indicators.

6. Document Findings

Four emails were analyzed as part of the phishing awareness assessment. Three emails displayed multiple phishing characteristics, including spoofed sender domains, urgent language, suspicious hyperlinks, and potentially malicious attachments. These emails were classified as **High** or **Critical** risk because they attempted to obtain sensitive information or deliver malware. The legitimate HR email originated from the organization's official domain and contained routine business communication without any phishing indicators.

7. Prevention & Awareness Guidelines

To protect against phishing attacks, users should follow these cybersecurity best practices:

- Verify that the sender's email address belongs to the official organization.
- Compare the sender's domain with the organization's genuine website before trusting the email.
- Hover over hyperlinks to inspect their destination before clicking.
- Avoid opening unexpected attachments, particularly ZIP, EXE, or macro-enabled Office documents.
- Never provide passwords, OTPs, banking credentials, or personal information through email.
- Be cautious of messages that create urgency or threaten immediate consequences.
- Verify unusual requests directly with the organization through official communication channels.
- Enable Multi-Factor Authentication (MFA) on important accounts.
- Keep operating systems, browsers, and antivirus software updated.
- Report suspected phishing emails to the organization's IT or security team immediately.

8. Actual Phishing emails Examples

Weekly Paid Job



Aisha S. Mansour <Aisha.Mansour@bot.go.tz>
To

I am sharing job opportunity information to individuals who might be interested in a paid UNICEF Part-Time job with a weekly paid job of \$500.00 USD.

If interested, Kindly contact Dr Thomas Nicholas via thomasnicholasdr@gmail.com N.B, this is strictly a work from home position.

Sign,
Career Opportunity

DISCLAIMER: Nothing contained in this e-mail, that is, its main text and the attachments thereto shall be construed as legally binding to the Bank of Tanzania.
Please, refer to the full disclaimer found at <http://www.bot.go.tz/EmailDisclaimer.asp>

